4. Derive “Currie Equation” with 99% confidence. For example, if somebody measured the no-source background and recorded 100 counts during a 2-min measurement, how should he/she set up the critical level and what is MDA with 99% confidence? We assume that the measurement time with an unknown source is also 2 mins.

2 The Single-Channel Analyzer

2.1 Equipment Setup

2.1.1 Required Equipment

- oscilloscope
- amplifier
- preamplifier
- single-channel analyzer (SCA)
- counter/timer
- high voltage power supply
- Geiger-Mueller (GM) tube
- thorium source

2.1.2 Measurement Setup

- Connect the pre-amplifier to the high voltage power supply and the G-M tube
- Connect the pre-amp to the amplifier, the amplifier to the SCA, and finally the SCA to the counter/timer
  - Split the amplifier output and SCA output to the oscillsocope.
- Set the power supply to + 800 V (**be careful when increasing the bias and do it slowly).**
  - **If your detector is labeled as something other than 800 V, only set the voltage to whatever it is labeled as**
- Place a radiation source (thorium source) on a shelf below the Geiger-Mueller (G-M) tube.
- Check the amplifier output on the scope and make sure you see pulses.
  - Set the amplifier to "Neg" since the GM tube produces negative pulses.
- Set the SCA to integral mode
- Find the LLD value that discriminates all pulses
- Reduce the LLD value until you see counts on the counter/timer
  - Make sure to keep the LLD high enough so that you're only triggering on detector events (no noise triggering).

2.2 Counting Measurements

2.2.1 Measurement of a Gaussian Distribution

- With the source in place, determine a shelf height and counting interval necessary to record counts with an average value of about 30.
  - Start with a lower shelf height to ensure the detector does not have high dead time.
  - It is better to have a lower count rate and higher count time to ensure the detector is not count-rate limited.
- Record a set of 50 of these counts. You may want to use a spreadsheet program to store your data. The dwell time on the counter can be used to adjust the time between different runs. From this data, calculate:

Experimental mean:

Sample variance:

2.2.2 Measurement of a Poisson Distribution

- Use the counter to accumulate 1000 consecutive trials from a source which will give an average of about 7 counts per timing interval. The raw data don't need to be included in the lab report.
  - To get a count rate of ~7, you will most likely need to use the lowest shelf height and a low counting interval.
  - You can decrease the "Dwell" time to reduce the time between samples.
- Tabulate the data by counting the number of times 0 counts have been stored in an interval, the number of times 1 has been stored in an interval, etc. up to the maximum number of counts of any interval (make a frequency table)
- Make a frequency plot of P(i) , the probability of occurrence of exactly “i” counts vs. the value of “i” from zero to the maximum